Comments from Reviewer 1

The main contribution of this paper is to show the low predictive power of sociodemographic characteristics on longevity. As discussed in the paper, this finding aligns with existing studies of the proportion of the variance explained by single SES variables based on cross-sectional life table analysis. Nevertheless, beyond confirming this previously known low explained variance in longevity, the paper is important because it alleviates two major concerns in the literature about why this might be: (1) it is not because of synthetic cohort assumptions or within-cohort mortality selection – this study shows that it also holds up for prospective monitoring of real cohorts; and (2) is not because a single SES variable is not capturing enough variation—even with multiple covariates mortality prediction is weak, simply because within-group variation in longevity is enormous. In other words, this paper confirms that mortality is highly stochastic.

I don't have any substantive concerns. The paper is beautifully written, and the frontend is especially thoughtful, integrating multiple strands of the literature. In fact, it was so well done that the back end of the paper felt a little disappointing by comparison (more on that later). The analysis is sound, and even if the data has some limitations (e.g. not capturing the early life course, limited variables, likely some linkage failures, generalizability across cohorts/contexts), the advantages of the cohort design likely outweigh the drawbacks. I don't see that any of these limitations would invalidate the main conclusions. With perfect detailed data covering the full lifespan of these cohorts it might be possible to triple or quadruple the predictive power of the covariates. Even still, predictability would be low, which I see as the main point.

I only have the following minor suggestions:

1. It could be stressed more that this paper is assessing the predictive power of a set of covariates among 30-year-olds, where the variation across individuals in their set of covariates might not be as meaningful for mortality as it would be at other life course stages. I'm not sure, but maybe there's a literature on this. It might also be that it's not so much the set of fixed covariates themselves that are predictive of mortality, but rather changes in certain of these covariates over the life course, which cannot be tested with this data but could be discussed. For example, if we could pick these individuals up at a later census, it's conceivable that changes in marital status, income, rurality, etc. could substantially improve mortality prediction. It's notable that marital status doesn't even show up among the 15 top predictors, possibly because marriage rates were high for this cohort and marital dissolution or early widowhood at a later age might be the more important mortality signal.

2. Keeping with the life course theme, I found Figure S6 to be interesting and surprising and worth further discussion, maybe even inclusion into the main text. I would have expected greater predictive power from observing cohorts at a later stage in their life course. This analysis finds the opposite, and I am wondering why this could be. Does it relate to the set of ages that are later observed? Other reasons?

3. On the one hand, the predictive power of one's set of covariates on longevity is low. On the other hand, between-group inequalities in mortality are not trivial. One finding that is coming out of the mortality

prediction literature is that covariates might have ok discriminatory power, to for example predict which of two individuals is likely to live longer (e.g. Bardolato et al, which you cite). Drawing on this literature, and its implications, would enrich the discussion, particularly around what importance we see more generally for the social determinants of mortality.

4. The ties to uncertainty demography is interesting and could be further developed. It's worth looking at the recent PDR paper by Sasson (2025), who makes a lot of these same arguments. The paper by Aburto et al. (2023) argues that the implications of lifetime uncertainty might be particularly salient in violent contexts.

5. On p.18 there are comparisons drawn to the predictability of other life events, from totally different datasets and study designs. While this is acknowledged – to me the designs are too incomparable for this statement: "lifespan is substantially more difficult to predict than material hardship at age 15 ($R2 = 0.23$) and is most similar to predicting primary caregiver layoff ($R2 = 0.03$)". The supplementary analysis itself showed that adding in age and looking at mortality prediction over a 5-year window substantially improved prediction.

Smaller things:

1. "Second, limited predictability reveals how inequality and uncertainty coexist: structural disparities shape the overall distribution of lifespans but remain limited in their ability to determine individual outcomes." – To me, this sentence goes against the idea that we can only explain a small portion of the variance. I would actually say that the distribution is defined by stochasticity but structural disparities shift and change the skewness.

2. First sentence of the background: "inequality in mortality has also increased over time" – This is probably true for the USA, but several European countries with long time series of data have experienced increasing and declining inequalities over time (Zazueta-Borboa et al., 2023), while other countries might not even be experiencing classic mortality gradients (Sudharsanan et al., 2020). It would be good to either rephrase this sentence or make it clear that you are referring to the USA in this section.

3. "Genetic, behavioral, or biomedical predictors fall outside the conceptual scope of our study, which is concerned with the social structuring of mortality." This is strongly worded, and only holds so long as these factors do not themselves predict the social statuses examined here.

4. What is the number of mothers/fathers/couples in Fig S1?

Finally, I think it's wonderful that the authors chose to make the analysis replicable in OSF.

Literature cited:

Aburto, J. M., Di Lego, V., Riffe, T., Kashyap, R., Van Raalte, A., & Torrisi, O. (2023). A global assessment of the impact of violence on lifetime uncertainty. Science advances, 9(5), eadd9038.

Sasson, I. (2025). A New Research Agenda for Social Inequalities in Mortality: Challenges and Open Questions. Population and Development Review, 51(1), 323-360.

Sudharsanan, N., Zhang, Y., Payne, C. F., Dow, W., & Crimmins, E. (2020). Education and adult mortality in middle-income countries: Surprising gradients in six nationally-representative longitudinal surveys. SSM-population health, 12, 100649.

Zazueta-Borboa, J. D., Martikainen, P., Aburto, J. M., Costa, G., Peltonen, R., Zengarini, N., Sizer, A., Kunst, A. E., & Janssen, F. (2023). Reversals in past long-term trends in educational inequalities in life expectancy for selected European countries. J Epidemiol Community Health, 77(7), 421-429.

Comments from Reviewer 2

This paper is an important contribution to the analysis of lifespan variation. The authors use linked individual records to associate individual lifespans with a large set of social variables, use machine learning to find a model relating lifespan to those variables, and report the fraction of variance (small) explained. The paper is extremely clearly written. It is a valuable contribution not only for the results of this particular study, but for the useful summary of previous work (thank you for Table 1!) on this topic.

I have a few thoughts to share. These are not criticisms, but may be helpful to the authors.

1. p. 2, line 19-20. I like this clear statement.

2. p. 3, line 36 and also p. 5, line 79. You say that decompositions are limited to a single categorical variate. This is no longer true. Caswell and van Daalen (2025, Vienna Yearbook of Population Research) show how to incorporate multiple factors and their interactions, including a couple of toy examples. A more complete analysis of a four-factor study will be presented at PAA 2026, and will be written up soon.

3. p. 4, line 47-49. Here, or perhaps in the discussion, you may want to mention stochasticity in other demographic outcomes in addition to longevity, including healthy longevity and lifetime reproductive output (LRO). Examples of these are shown in Caswell (2023), which you cite. An example of partition of variance in LRO is given in van Daalen et al. (2022, American Naturalist) for a laboratory experiment on aging in a rotifer. My point is just to emphasize that the argument of Trinitapoli that variance is an important component of analysis, not just error, applies beyond analysis of lifespan.

4. p. 5, footnote 2. This is an important point. There is a difference between variance obtained from actual observations of individual lifetimes (which you do) and variance calculated from the rates estimated for groups (which most decompositions do). [We need a good terminology to discuss these.] In the former case, the variance due to stochasticity may indeed be due to unmeasured heterogeneity. In the latter case, it cannot, because the calculation of variance, from a life table or other Markov chain,

explicitly treats every individual within a group as identical. (I discuss this a little on the top of p. 328 of Caswell 2023). I have always said that data on individual lives are (sometimes) available to population biologists, but not to demographers, because we can't put leg bands on people when they are born like we do with albatrosses. But this study, and studies using register data, are changing that.

5. p. 8, line 128. A set of cohort results for variance due to frailty can be found in Hartemink et al. (2017, Theoretical Population Biology 114:107-116), for France, Sweden and Italy.

6. p. 10, line 163. What does "normalized" mean here?

7. Figure 3 and elsewhere. This is a nice way to display these results.

In sum, this paper makes a significant contribution to the literature on variation in longevity. I enjoyed reading it.